HR DATA ANALYSIS PROJECT REOPRT

Project Overview:

Received 2 years' data from HR department of a company with 1480 records, data were largely clean except few preprocessing was to be done. After understanding data and done few basic cleaning such as removing duplicate records and handling missing values in power query, developed interactive dashboard and found important insights, suggested it to HR for data driven actions.

Problem Statement:

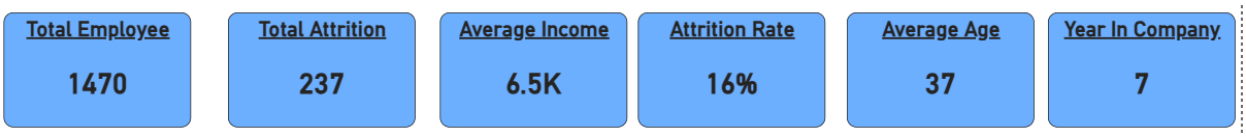
HR department were experiencing high attrition. I was assigned to data driven suggestions to find root cause and minimize attrition.

Tools Used:

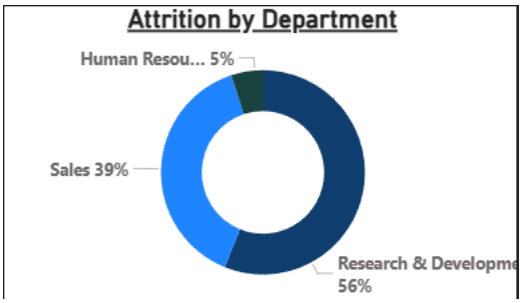
- Excel
- Power Query
- Power BI

Procedure:

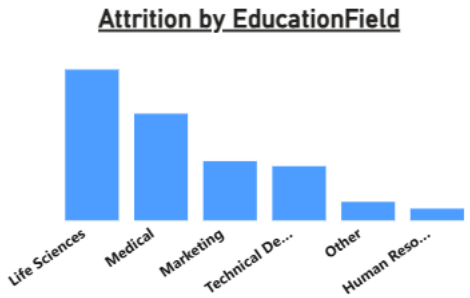
1) Dataset was imported to power query; Employee ID column contain duplicate records. Deleted entire rows containing duplicate records.
2) Attrition in text format containing 'yes', 'no' as our main focus was to calculate attrition which was not possible in text format, added conditional column where 'yes' was assigned as '1' and 'no' as '0', and converted its data type to whole number.
3) 'Years with current manager' column contained 4% blank values, its' weak correlations with attrition this column is not useful, deleted column.
4) Cleaned data were imported to power query to prepare interactive dashboard.
5) Prepared KPI using dax to summarize dataset.



6) Analyzed attrition by department using a donut chart, identifying Research & Development as the department with the highest attrition (56% of total).

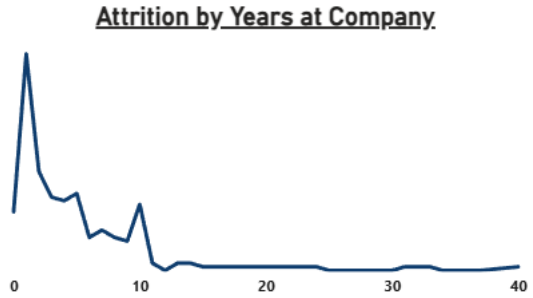


7) Using clustered column charts analyzed attrition by education field. Life Sciences had highest attrition of 89 that counts 37.55% of total attrition, whereas HR count lowest of 7 that's 2.95%

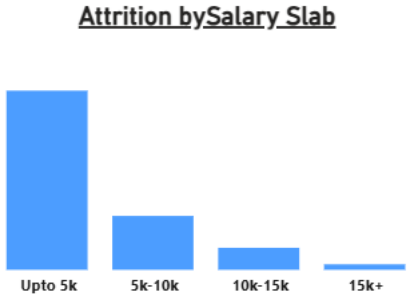


8) Using clustered bar chart analyzed attrition count by job role. Laboratory technician and sales executive were top 2 role of attrition, top 2 carry 50.21% and research director is among least 0.84%.

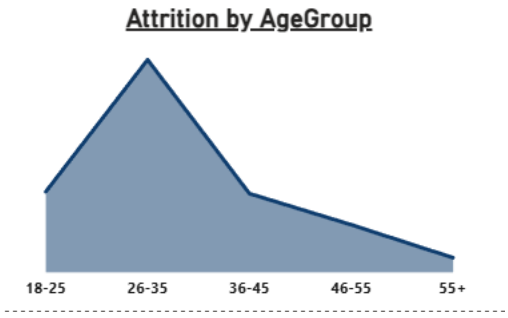
9) Line chart was used to analyze attrition by tenure. Graph shows that most attritions are at early years (0-2 years) of working with company, 9th and 10th year shows noticeable spike, after that attrition almost were minimal.



10) Used clustered column chart to analyze attrition by salary slab. Salary slab of up to 5K had most attrition that account 68.78% of total, attrition decrease as salary slab increase.



11) Analyzed attrition by age group by using area chart. Attrition was high among age 18-45, it accounts 85.65% of total. Attrition decreased after 45 as age increased.



12) Used dropdown slicer to filter and analyzed all the kpi and charts in gender basis.

Key Findings:

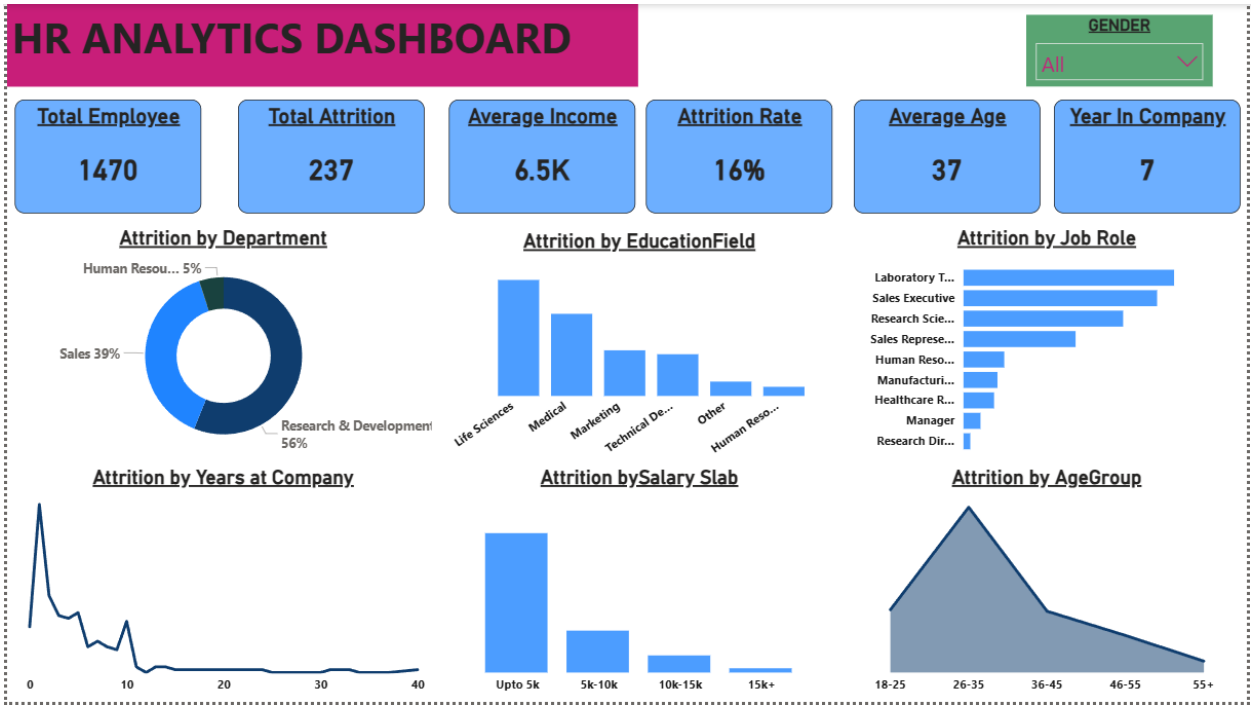
- Attrition of males were high accounts 63.29% while women were on 36.71%
- Among total attritions 95% were from R&D and sales department. This is alarming trend as these two departments are key to run any organizations.
- Employees prefer to leave it on early stages (0-2 years), and salary slab up to 5K were more likely to move on, this highlights we were not able offer good work culture and enough salary according to job roles. It clearly indicates, to reduce attrition we need to focus on welfares of employees and review entry level salary according to competitive market.
- Core workforces (18-45 years) are leaving company in high numbers, this clearly show that we were not able to show them clear pictures of career growth. HR should implement structured, department-specific career progression frameworks, clearly defining growth paths, skill requirements, and promotion timelines for each role.
- Laboratory technician, sales executive, research scientist was among top roles by job attrition indicate that we are losing workforce after spending good resources on them for training, and upskills. This represents a direct cost and productivity loss to the organization.

Suggestions:

16% Attrition is very high for company to have sustainable growth. After analyzing data here are my suggestions for the actions on department wise:

Finance:

- Allocate enough budget to HR for hiring and retention as insufficient salary is main reason of attrition.
- Until formal salary hike revision, arrange allowance as immediate measure.

HR:

- Ensures clear roadmap is prepared on career growth of employees from all departments.
- Review overall policy and take corrective measures on employee welfare program.
- Review company's policy on learning and development. Plan trainings on overall professionalism and personal growths of employee.
- Consult with existing employees especially (18-45 years) age group about their expectation and goals, it will give clear picture and will help to lower communication gap.